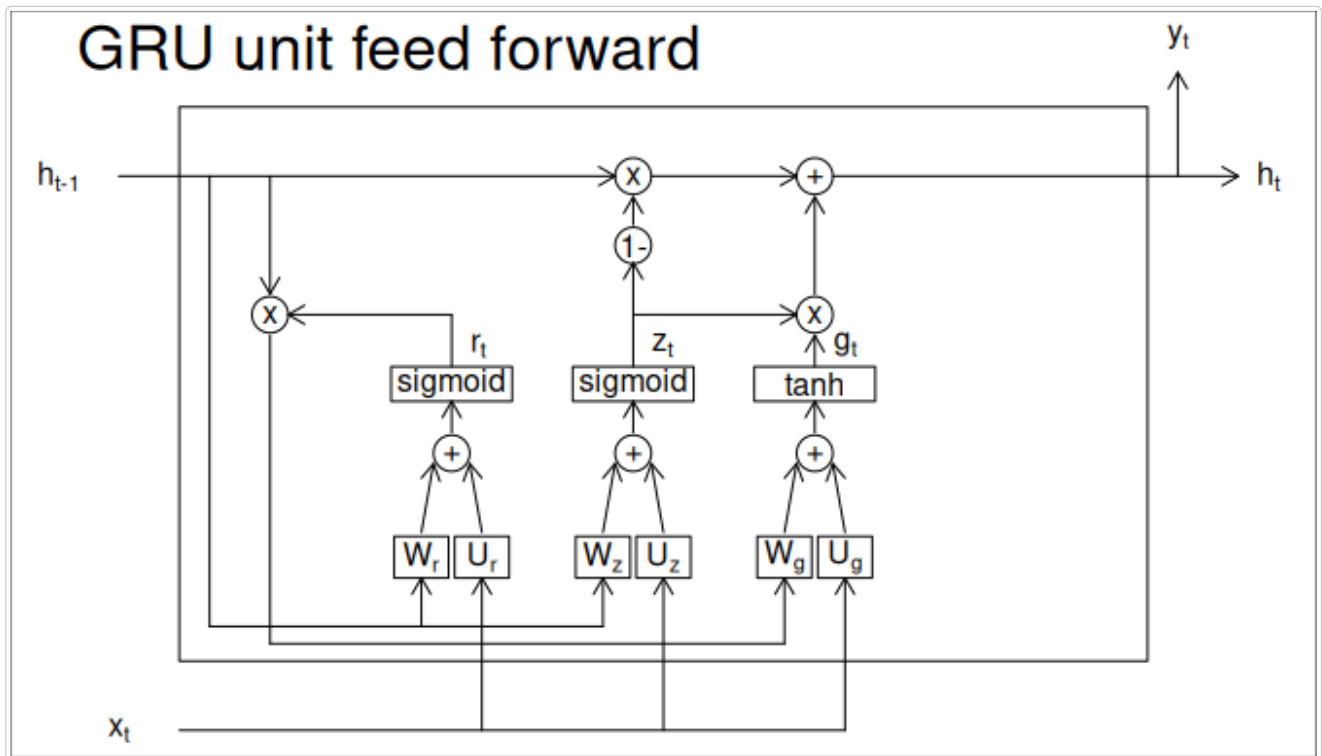


GRU units

Dimitri Fichou

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Feed forward pass



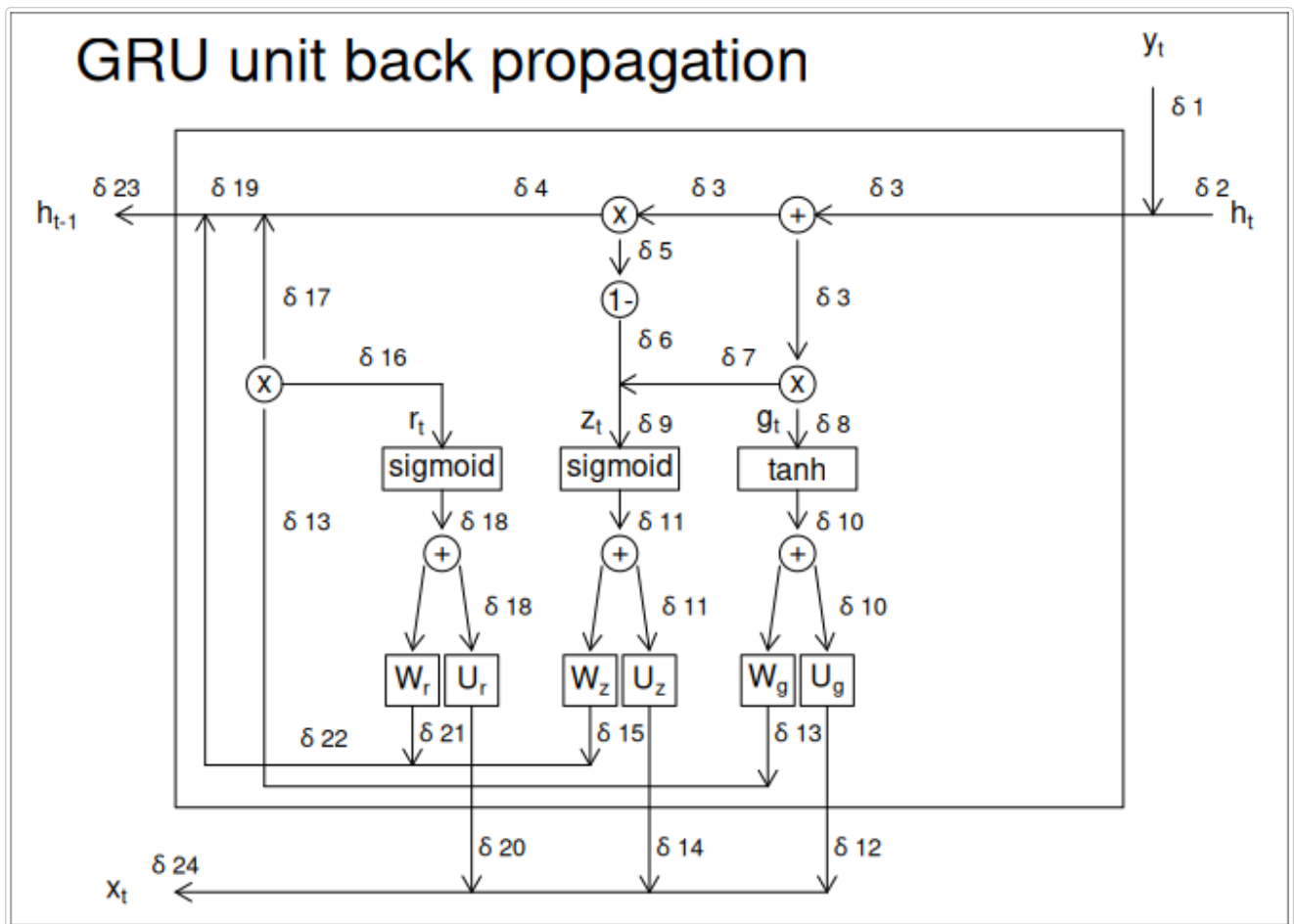
$$r_t = \text{sigmoid}(i \cdot W_r + x_t \cdot U_r)$$

$$z_t = \text{sigmoid}((h_{t-1} \cdot W_z + x_t \cdot U_z)$$

$$g_t = \tanh(W_g \cdot (h_{t-1} \cdot r_t) + x_t \cdot U_g)$$

$$h_t = y_t \oplus h_{t-1} \cdot ((1 - z_t) + (z_t) \cdot g_t)$$

Back propagation pass



To perform the BPTT with a GRU unit, we have the error coming from the top layer (d_1), the hidden future states (d_2). Also, we have been stored during the feed forward the states at each step of the feeding. In the case of the future layer, this error is just set to zero if not yet calculated. For convention, $\cdot$ correspond to point wise multiplication, while $*$ correspond to matrix multiplication.

The rules on how to back propagate come from [this post](#).

$$d_3 = d_1 + d_2$$

$$d_4 = (1 - z_t) \cdot d_3$$

$$d_5 = d_3 \cdot h_{t-1}$$

$$d_6 = 1 - d_5$$

$$d_7 = d_3 \cdot g_t$$

$$d_8 = d_3 \cdot z_t$$

$$d_9 = d_7 + d_8$$

$$d_{10} = d_8 \cdot \tanh(g_t)$$

$$d_{11} = d_9 \cdot \text{sigmoid}(z_t)$$

$$d_{12} = d_{10} * W_g^T$$

$$d_{13} = d_{10} * U_g^T$$

$$d_{14} = d_{11} * W_z^T$$

$$d_{15} = d_{11} * U_z^T$$

$$d_{16} = d_{13} \cdot h_{t-1}$$

$$d_{17} = d_{13} \cdot r_t$$

$$d_{18} = d_{17} \cdot \text{sigmoid}'(r_t)$$

$$d_{19} = d_{17} + d_4$$

$$d_{20} = d_{18} * W_r^T$$

$$\delta d_{21} = d_{18} * U_r^T$$

$$d_{22} = d_{21} + d_{15}$$

$$d_{23} = d_{19} + d_{22}$$

$$d_{24} = d_{12} + d_{14} + d_{20}$$

The error d_{23} and d_{24} are for the next layers. Once all those are available, it is able to calculate the update weight.

$$dW_r = \delta W_f + h_{t-1}^t * d_{10}$$

$$\delta U_r = \delta U_f + x_t^T * d_{10}$$

$$dW_z = \delta W_i + h_{t-1}^t * d_{11}$$

$$dU_z = \delta U_i + x_t^T * d_{11}$$

$$dW_g = \delta W_+ (h_{t-1}^T r_t) * d_{18}$$

$$dU_{\bar{g}} = \delta U_g + x_t^T * d_{18}$$